

Vulnerability Assessment Report

Cyber Security Internship – Future Interns

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Tools Used: Nmap, OWASP ZAP

Date: 15 May 2026

1. Introduction

This report presents a basic vulnerability assessment performed on a web application using Nmap and OWASP ZAP. The purpose of the assessment was to identify common security weaknesses and provide remediation recommendations.

2. Scope

The assessment focused on analyzing publicly accessible web application components and identifying common security issues through passive scanning techniques.

3. Tools Used

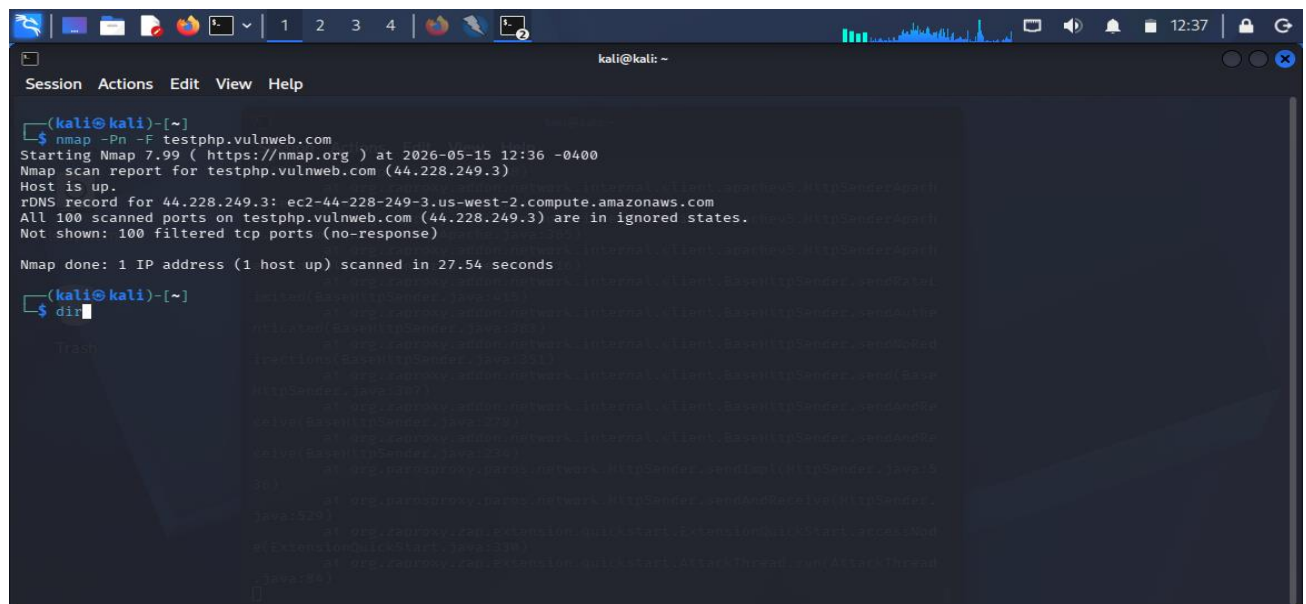
- Nmap
- OWASP ZAP
- Browser DevTools

4. Methodology

The assessment included reconnaissance using Nmap and passive web vulnerability analysis using OWASP ZAP.

5. Reconnaissance Findings

Nmap scan results indicated that the target host was reachable but protected by filtering mechanisms that blocked visibility into common TCP ports.

A screenshot of a Kali Linux terminal window. The terminal shows the execution of an Nmap scan command: `nmap -Pn -F testphp.vulnweb.com`. The output indicates that the host is up and provides a rDNS record: `ec2-44-228-249-3.us-west-2.compute.amazonaws.com`. It also states that all 100 scanned ports are in ignored states and that 100 filtered TCP ports were not shown due to no response. The scan was completed in 27.54 seconds. The terminal window has a dark theme and standard window controls at the top.

- Figure 1: Nmap scan results showing the target host is reachable while TCP ports are protected by filtering mechanisms.

6. Vulnerability Findings

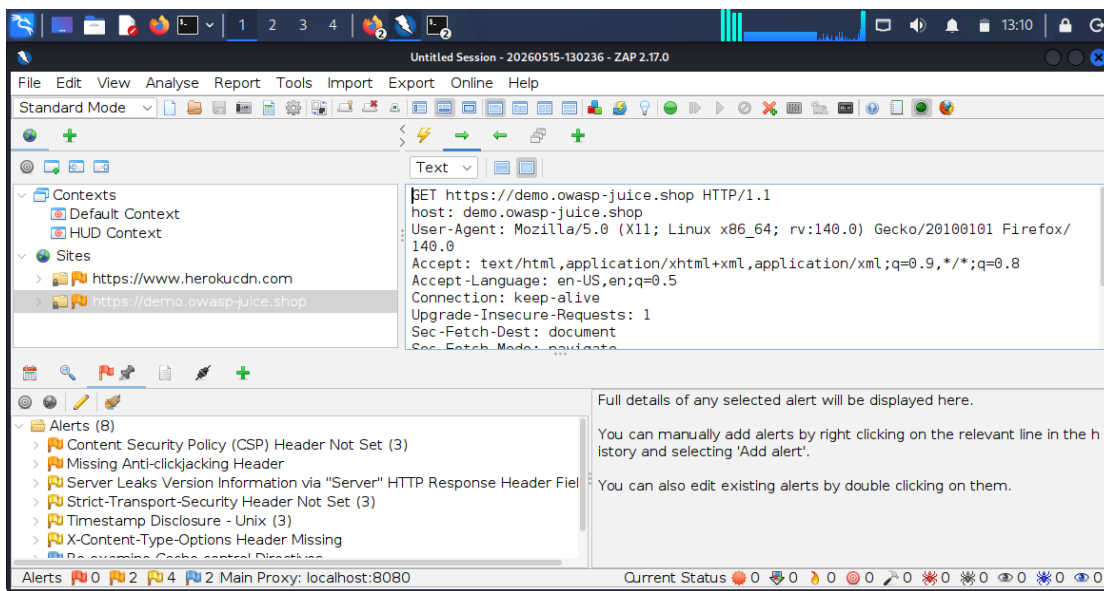
Example Findings:

1. Missing Content Security Policy Header – Medium Risk

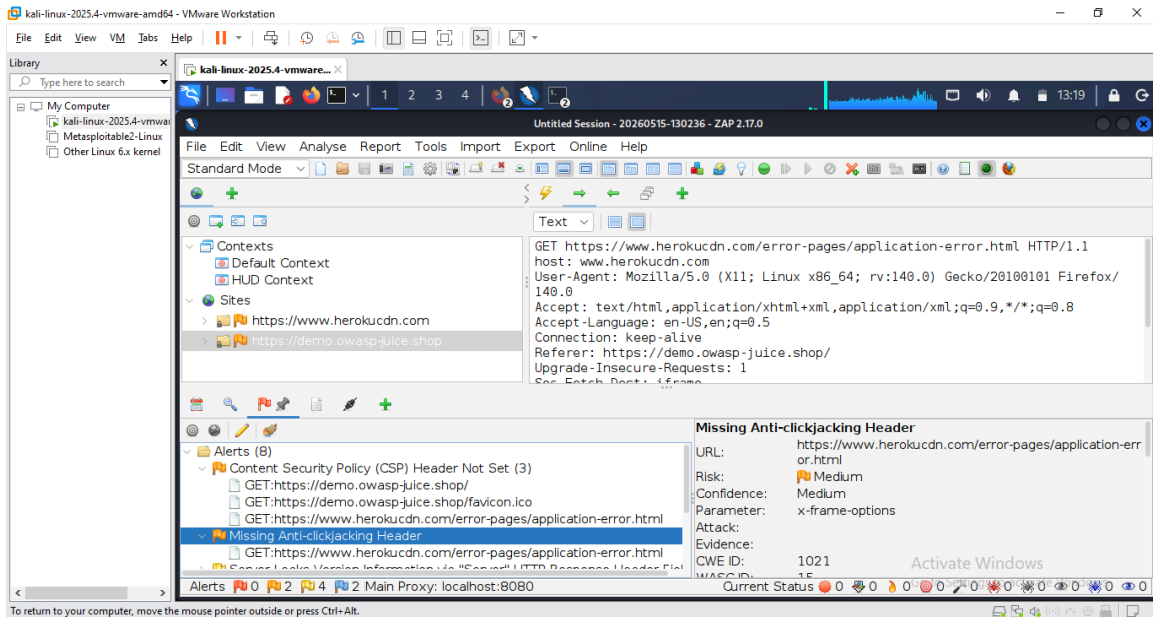
Recommendation: Configure proper Content-Security-Policy headers.

2. Cookie Without SameSite Attribute – Low Risk

Recommendation: Add SameSite attribute to cookies.



- Figure 2: OWASP ZAP successfully connected to the target web application for passive security analysis.



- Figure 3: OWASP ZAP alerts panel displaying identified security issues and associated risk levels.

7. Recommendations

Implement secure HTTP headers, improve cookie security, and reduce unnecessary information disclosure to improve the security posture of the application.

8. Conclusion

The assessment identified multiple low and medium-risk issues. Addressing these findings can significantly improve the security and resilience of the web application.

GitHub Repository Name: **FUTURE_CS_01**